WENQI CUI

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PROFESSIONAL EMPLOYMENT

New York University, New York, NY *Sept. 2025 -*
Clare Boothe Luce Assistant Professor, Department of Electrical and Computer Engineering

California Institute of Technology, Pasadena, CA *July. 2024 - Aug. 2025*
Postdoctoral Scholar, Computing + Mathematical Sciences
Advisor: Prof. Steven Low, Prof. Adam Wierman

EDUCATION

University of Washington, Seattle, WA *Sept. 2019 - Jun. 2024*
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Baosen Zhang

Zhejiang University, Hangzhou, China *Sept. 2016 - Jun. 2019*
Master of Science in Electrical Engineering
Advisor: Prof. Yi Ding

Southeast University, Nanjing, China *Aug. 2012 - Jun. 2016*
Bachelor of Engineering in Electrical Engineering and Automation

INDUSTRY EXPERIENCE

Microsoft Research, Redmond, WA *Jun.2022 -Sept.2022*
Research Intern at Microsoft Research Special Projects at Remond Lab Mentor: Weiwei Yang
We investigated efficient reinforcement learning algorithms for controlling large-scale physical systems, including power systems and traffic networks. The algorithms seek to overcome the challenges of partial observability, sample complexity, and distribution shift of trajectories from updated control policies.

Microsoft Research, Redmond, WA *Jun.2021 -Sept.2021*
Research Intern at Microsoft Research Special Projects at Remond Lab Mentor: Weiwei Yang
We proposed a novel framework for predicting power system dynamics and transients in the frequency domain, which provides a computation speed up of more than 400 times compared to existing power system tools. This work also led to the acquisition of a grant from "Climate Change AI Innovation Grants Program" .

PUBLICATIONS

Preprints

[P1]. J. Wang, J. Singh, H. Xu, X. Li, M. Yan, S. Liu, Y. Li, Y. Chen, **W. Cui**, "LLM-PowerTrace: A Dataset of Time-Resolved GPU Power Traces for LLM Fine-Tuning and Inference", submitted to *International Conference on Learning Representations (ICLR)*, 2026.

[P2]. S. R. Singh, **W. Cui**, "Learning Lyapunov Functions for Power System Transient Stability: Neural Verification and Retraining", accepted to *Tackling Climate Change with Machine Learning Workshop at NeurIPS*, 2026.

[P3]. M. Yan, H. Wang, L. Biao, H. J. Chao, S. Q. Zhang, **W. Cui**, "Analytical Power-Aware Provisioning for Prefill–Decode Disaggregated AI Inference", submitted to *ACM e-Energy Conference*, 2026.

[P4]. M. Siddiqui, D. Rojas, C. Yang, **W. Cui**, Y. Shi, Y. Chen, "Measured Joules, Learned Routes: Learning to Route for Energy-Efficient LLM Serving", submitted to *ACM e-Energy Conference*, 2026.

[P5]. Y. Dong, **W. Cui**, "Decentralized Gain Learning for Voltage Control via Local Trajectory Convolution", submitted to *American Control Conference (ACC)*, 2026.

[P6]. W. Wang, A. Bhatnagar, **W. Cui**, I. Incer, A. Chen "Synthesizing Voltage Ride-Through Controllers for Data Centers", accepted to *ACM Conference on Digital Transformation*, 2026.

[P7]. M. Yan, Trager. Jones, B. Zhang, Y. Chen, **W. Cui**, "Switching-Reference Voltage Control for Distribution Systems with AI-Training Data Centers", submitted to *IEEE Transactions on Smart Grid*, 2026.

[P8]. Y. Dong, **W. Cui**, Han Xu, A. Wierman, and S. Low, "Data-Driven Successive Linearization for Optimal Voltage Control", submitted to *IEEE Transactions on Control of Networked Systems*, 2026.

[P9]. **W. Cui**, Y. Xie, S. Low, A. Wierman and B.Zhang, "Leveraging Predictions in Power System Voltage Control: An Adaptive Approach", submitted to *IEEE Transactions on Smart Grid*, 2025.

Journal Papers

[J1]. X. Han, Y. Xie, **W. Cui**, T. Li, S. Low, "Robust Solar Hosting Capacity for Pasadena's Decarbonization", *IEEE Energy Sustainability Magazine*, 2025.

[J2]. J.Feng, **W. Cui**, J.Cortés and Y.Shi, "Online Event-Triggered Switching for Frequency Control in Power Grids with Variable Inertia", *IEEE Transactions on Power Systems*, 2025.

[J3]. **W. Cui**, Y. Jiang and B. Zhang, "Reinforcement Learning for Optimal Frequency Control: A Lyapunov Approach", *IEEE Transactions on Power Systems*, vol. 38, no. 2, pp. 1676-1688, 2023.

[J4]. **W. Cui**, W. Yang and B. Zhang, "A Frequency Domain Approach to Predict Power System Transients", *IEEE Transactions on Power Systems,* 2023.

[J5]. J.Feng, **W. Cui**, J.Cortés and Y.Shi, "Bridging Transient and Steady-State Performance in Voltage Control: A Reinforcement Learning Approach with Safe Gradient Flow", *IEEE Control Systems Letters (L-CSS)*, vol. 7, pp. 2845-2850, 2023.

[J6]. **W. Cui** and B. Zhang, "Equilibrium-Independent Stability Analysis for Distribution Systems with Lossy Transmission Lines ", *IEEE Control Systems Letters (L-CSS)*, vol. 6, pp. 3349-3354, 2022.

[J7]. **W. Cui**, J. Li and B. Zhang, "Decentralized Safe Reinforcement Learning for Inverter-Based Voltage Control", *Electric Power Systems Research [journal version of Power Systems Computation Conference (PSCC)]*, vol. 211, p. 108609, 2022.

[J8]. **W. Cui** and B. Zhang, "Lyapunov-Regularized Reinforcement Learning for Power System Transient Stability ", *IEEE Control Systems Letters (L-CSS)*, vol. 6, pp. 974-979, 2022.

[J9]. Y. Jiang, **W. Cui**, B. Zhang and J.Cortés, "Stable Reinforcement Learning for Optimal Frequency Control: A Distributed Averaging-Based Integral Approach", *IEEE Open Journal of Control Systems*, vol. 1, pp. 194-209, 2022.

[J10]. C. Doty*, S. Gallagher*, **W. Cui***, W. Chen*, S. Bhushan*, M. Oostrom, S. Akers, S. Spurgeon, "Design of a Graphical User Interface for Few-Shot Machine Learning-Based Classification of Electron Microscopy Data", *Computational Materials Science*, vol. 203, p. 111121, 2022. (* authors contributed equally)

[J11]. N. Shang, Y. Ding, **W. Cui**, "Review of Market Power Assessment and Mitigation In the Reshaping of Power Systems: State-of-Art Status and Potential Research Studies", *Journal of Modern Power System and Clean Energy*, vol. 10, no. 5, pp. 1067-1084, 2022.

[J12]. Y. Ding, **W. Cui**, S. Zhang, H. Hui, Y. Qiu, Y. Song, "Multi-State Operating Reserve Model of Aggregate Thermostatically-Controlled-Loads for Power System Short-Term Reliability Evaluation", *Applied Energy*, vol. 241, pp. 46-58, 2019.

[J13]. **W. Cui**, Y. Ding, H. Hui, Z. Lin, P. Du, Y. Song, C. Shao, "Evaluation and Sequential-Dispatch of Operating Reserve Provided by Air Conditioners Considering Lead-Lag Rebound Effect", *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6935-6950, 2018.

Conference Papers

[C1]. Y. Xie*, **W. Cui***, A. Wierman, "Enhancing Data Center Low-Voltage Ride-Through", *ACM SIGEnergy e-Energy Conference*, 2026. (* authors contributed equally)

[C2]. N.Christianson, **W. Cui**, S. Low, W. Yang and B.Zhang, "Fast and Reliable $N-k$ Contingency Screening with Input-Convex Neural Networks", *Learning for Dynamics & Control Conference,* 2025.

[C3]. **W. Cui**, Y. Jiang, B. Zhang and Y.Shi, "Structured Neural-PI Control with End-to-End Stability and Output Tracking Guarantees", *Conference on Neural Information Processing Systems (NeurIPS)*, 2023.

[C4]. **W. Cui**, G. Shi, Y.Shi and B. Zhang, "Leveraging Predictions in Power System Frequency Control: an Adaptive Approach", *IEEE Conference on Decision and Control*, 2023.

[C5]. **W. Cui**, L. Huang, W. Yang and B. Zhang, "Efficient Reinforcement Learning Through Trajectory Generation", *Learning for Dynamics & Control Conference,* 2023.

[C6]. Y. Jiang, **W. Cui**, B. Zhang and J.Cortés, "Equilibria of Fully Decentralized Learning in Networked Systems", *Learning for Dynamics & Control Conference,* 2023.

INVITED TALKS AND PRESENTATIONS

[1]. "Adaptive Voltage Control Under Rapid Net-Load Fluctuations in Distribution Systems ", Massachusetts Institute of Technology, 2026/3.

[2]. "Data-Driven Voltage Control for Power Distribution Systems ", Resnick Sustainability Institute, 2025/12.

[3]. "Structured Control and Learning for Networked Systems ", University of California Merced, 2025/11.

[4]. "Structured Control and Learning for Networked Systems ", University of Michigan, 2025/10.

[5]. "Leveraging Predictions in Power System Voltage Control: an Adaptive Approach", International Conference on Continuous Optimization, 2025/06.

[6]. "Structured Control and Learning in Energy Systems", Department of Mathematics, Florida State University, 2024/11.

[7]. "Leveraging Predictions in Power System Frequency Control: an Adaptive Approach ", 2024 INFORMS Annual Meeting , Seattle, 2024/10.

[8]. "Structured Control and Learning in Energy Systems", Department of Electrical and Computer Engineering, University of California San Diego, 2023/06.

[9]. "Efficient Reinforcement Learning Through Trajectory Generation", Learning for Dynamics & Control Conference, University of Pennsylvania, 2023/06.

[10]. "Structured Neural-PI Control for Networked Systems: Stability and Steady-State Optimality Guarantees", Department of Aeronautics and Astronautics, University of Washington, 2023/05.

[11]. "Equilibrium-Independent Stability Analysis for Distribution Systems with Lossy Transmission Lines ", the 61th IEEE conference on Decision and Control, Cancun, Mexico, 2022/12.

[12]. "Structured Neural-PI Control for Networked Systems: Stability and Steady-State Optimality Guarantees", AI Power Lunch, Microsoft Research, Redmond, WA, 2022/08.

[13]. "Decentralized Safe Reinforcement Learning for Voltage Control", Power Systems Computation Conference, Porto, Portugal, 2022/06.

[14]. "Predicting Power System Dynamics and Transients: A Frequency Domain Approach", Azure Global Commercial Industry, Microsoft, Redmond, WA, 2021/09.

[15]. "Safe Reinforcement Learning for Optimal Frequency Control", Department of Electrical and Computer Engineering, University of Texas at Austin, 2021/08.

[16]. "Reinforcement Learning for Optimal Frequency Control: A Lyapunov Approach", *Tackling Climate Change with Machine Learning workshop at ICML 2021*, spotlight talk, 2021/07.

TEACHING EXPERIENCE

Lecturer for ECE 3143 Introduction to Smart Grids, New York University	*Sept.2026 -Dec.2026*
Lecturer for ECE 6653 Power System Operation and Control, New York University	*Jan.2026 -May.2026*

STUDENT ADVISING

Yiwei Dong, PhD student	*Sept.2025-*
Mingyuan Yan, PhD student	*Sept.2025-*
Shubham Ojha, Master student	*May.2026-*
Jiansheng Wang, Master student	*Jan.2026-*
Haochen Xu, Undergraduate student	*Jan.2026-*
Xinyi Li, Undergraduate student	*May.2026-*
Sehaj Randhir Singh, High school student (Queens High School For The Sciences)	*May.2026-*

HONORS & AWARDS

2025	Finalist for Outstanding Doctoral Dissertation, IEEE Power and Energy Society
2024	Pioneer Postdoctoral Fellowship, California Institute of Technology
2024	PIMCO Postdoc Fellowship, California Institute of Technology
2023	Rising Stars in Cyber-Physical Systems (CPS), University of Virginia
2023	Best Poster Award, Grid Science Winter School & Conference, Los Alamos National Lab
2022	Rising Stars in EECS, University of Texas at Austin
2021	Sarala Vadari Award, University of Washington
2020	Clean Energy Institute (CEI) Fellowship, University of Washington
2019	Rushmer Innovator Fellowship, University of Washington
2018 & 2014	National Scholarship, Chinese Ministry of Education

SERVICES & ACTIVITIES

Technical Program Committee: ACM e-Energy 2026, ACM DXConf 2026

Workshop organizer: NeurIPS 2026 Workshop on AI Foundations for Power Systems; Workshop on Physics-Informed Learning for Energy Systems at ACM e-Energy 2026

Journal & Conference Reviewer: IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Automatic Control, IEEE Transactions on Control of Networked Systems, IEEE Control Systems Letters, ACC, CDC, AAAI, etc.